ivcheck: Tests for Instrumental Variable Validity in R

by Charles Coverdale

Abstract The **ivcheck** package implements three falsification tests of the identifying assumptions behind instrumental variable estimation of the local average treatment effect: Kitagawa (2015), Mourifié and Wan (2017), and Frandsen, Lefgren, and Leslie (2023). Each test had existed only as author-hosted replication code in Stata or Matlab prior to this package. **ivcheck** exposes each as a named R function with S3 methods for fitted **fixest** and **ivreg** models, plus a one-shot wrapper that runs every applicable test on a fitted model in a single call. The implementation uses the variance- weighted Kolmogorov-Smirnov form of Kitagawa, the full Chernozhukov- Lee-Rosen intersection-bounds inference with Andrews-Soares adaptive moment selection for Mourifié-Wan, and a chi-squared linearity specialisation of the Frandsen-Lefgren-Leslie overidentification statistic (with a natural extension to multivalued treatment); the bounded-slope inequality component of the full Frandsen- Lefgren-Leslie test is deferred to a future release. Monte Carlo validation confirms that the empirical null distributions track the asymptotic reference distributions closely, with the judge-fixed-effects test mildly conservative at modest per-cell sample sizes.

1 Introduction

The **ivcheck** package provides falsification tests of the identifying assumptions behind instrumental variable (IV) estimation of the local average treatment effect (LATE). It implements three tests from the econometric literature (Kitagawa, 2015; Mourifié and Wan, 2017; Frandsen et al., 2023), each with S3 methods for fitted **fixest** and **ivreg** model objects, plus a wrapper that runs every applicable test on a single fitted model.

Applied IV practice has a tooling gap. The identifying assumptions of the Imbens and Angrist (1994) LATE framework, the local exclusion restriction and monotonicity, are routinely defended by hand-wave rather than tested, despite each having testable implications derived in the methodological literature since 2015. The reason is not conviction but distribution: Kitagawa's (2015) test was released with supplementary Matlab code on the author's website; Mourifié and Wan's (2017) reformulation relies on the Chernozhukov-Lee-Rosen (2013) `clrtest` Stata module; Frandsen, Lefgren, and Leslie (2023) ship a Stata SSC module `testjfe`. To our knowledge no package implements all three tests, and none of the three is available on CRAN. The closest prior work is the GitHub-hosted package **TestforInstrumentValidity** accompanying Carr and Kitagawa (2021), which implements the Kitagawa (2015) test with covariates but is not on CRAN and does not cover the Mourifié-Wan or Frandsen-Lefgren-Leslie tests. **ivcheck** is the first CRAN package in this family and the first to cover all three. The adjacent **ivDiag** (Lal et al., 2024) covers weak-instrument diagnostics (effective F, Anderson-Rubin, valid-t, local-to-zero) but does not implement LATE-validity tests; the R ecosystem has lacked a counterpart to that package for the falsification-test family.

ivcheck closes that gap. The package has three functional commitments. First, each test is a named R function (`iv_kitagawa`, `iv_mw`, `iv_testjfe`) that accepts either raw `(y, d, z)` vectors or a fitted IV model through S3 dispatch. Second, the implementations follow the published forms: Kitagawa's variance-weighted Kolmogorov-Smirnov statistic from Section 2.1 of that paper (equation 2.1), the Chernozhukov-Lee-Rosen intersection-bounds inference with Andrews-Soares (2010) adaptive moment selection for the conditional Mourifié-Wan test, and a chi-squared linearity specialisation of the Frandsen-Lefgren-Leslie overidentification statistic (a Sargan-Hansen special case of their regression test) with a natural extension to multivalued treatment. Third, a one-shot `iv_check` wrapper detects which tests are applicable from the structure of a fitted IV model and runs them, producing a tidy summary ready for inclusion in a paper's appendix.

2 Background

2.1 The LATE framework and its identifying assumptions

The Imbens-Angrist (1994) LATE framework identifies the causal effect of a binary treatment `D` on an outcome `Y` for the subpopulation of compliers under two untestable-looking assumptions about an instrument `Z`: the local exclusion restriction (`Z` affects `Y` only through `D`) and monotonicity (no unit's treatment assignment flips in the opposite direction to the average response to `Z`). Together with independence, these identify the LATE.

Function	Test or role	Key arguments
`iv_kitagawa()`	Kitagawa (2015) variance-weighted Kolmogorov-Smirnov test; multivalued treatment via Sun (2023)	`weighting`, `se_floor`, `n_boot`, `multiplier`, `treatment_order`
`iv_mw()`	Mourifié-Wan (2017) conditional moment-inequality test (Chernozhukov-Lee-Rosen bounds)	`x`, `basis_order`, `x_grid_size`, `y_grid_size`, `adaptive`, `n_boot`
`iv_testjfe()`	Frandsen-Lefgren-Leslie (2023) judge/group-IV overidentification test	`method`, `basis_order`, `n_boot`
`iv_check()`	Wrapper: detects and runs the applicable tests on a fitted IV model	`tests`, `alpha`, `n_boot`
`iv_power()`	Monte Carlo power and size curves for design planning	`method`, `n_sims`, `delta_grid`, `n_boot`

Table 1: Exported functions of **ivcheck**. The three test functions are S3 generics with methods for **fixest**, **ivreg**, and raw `(y, d, z)` vectors.

2.2 Testable implications

Under LATE identification, the joint distribution of `(Y, D, Z)` is constrained. Kitagawa (2015) shows that the population conditional joint CDFs `F(y, d | z) = Pr(Y <= y, D = d | Z = z)` must satisfy a set of stochastic dominance inequalities indexed by `(y, d)`, and that these inequalities are sharp characterisations of the identifying assumptions. Mourifié and Wan (2017) reformulates the implications as conditional moment inequalities that can accept covariates `X` through the Chernozhukov-Lee-Rosen (2013) intersection-bounds framework. Frandsen et al. (2023) derive a dedicated test for the judge-fixed-effects design where the instrument is a set of mutually exclusive dummies: under the joint null, the per-judge outcome mean `mu_j = E[Y | J = j]` is a linear function of the per-judge propensity vector `P_j`.

All three tests share a structural feature: they test necessary conditions, not sufficient ones. Failure to reject is evidence of no detectable violation at level `alpha`; rejection is evidence that at least one of exclusion or monotonicity has failed. The tests do not localise which assumption failed.

3 Package design

Table 1 summarises the package's five exported functions: three falsification tests, a wrapper, and a power calculator.

3.1 Architecture and dependencies

ivcheck is a pure-computation package. No network access is performed at runtime. R version 4.1.0 or later is required. Hard imports are `cli`, `stats`, and `parallel`; the bootstrap uses `parallel::mclapply` on POSIX systems with an explicit 2-core cap under R's `_R_CHECK_LIMIT_CORES_` environment variable. **fixest**, **ivreg**, **modelsummary**, and **broom** are in Suggests and are conditionally registered at load time via `.onLoad`.

3.2 The `iv_test` class

Every test returns an object of class `iv_test`, a list with uniform slots: the test statistic, the bootstrap or asymptotic p-value, the significance level, the bootstrap statistics vector, a `binding` list identifying the `(z, z', d, y)` configuration of the observed statistic, and the sample size. Test-specific slots carry additional information: `weighting` for `iv_kitagawa`; `conditional` (a flag for whether a covariate was supplied) and `kappa_n` for `iv_mw`; `pairwise_late` and `worst_pair` for `iv_testjfe`. A `print.iv_test` method emits a banner with the test name followed by three content lines (sample size, statistic and p-value, verdict); `plot.iv_test` shows the bootstrap distribution with the observed statistic marked.

3.3 S3 dispatch on fitted IV models

Each test function is an S3 generic with methods for `fixest`, `ivreg`, and the default numeric-vector interface. The `fixest` method recovers `y` via `model.matrix(m, type = "lhs")`, `d` via `fitted(fs) +`

`residuals(fs)` from the first-stage fit, and `z` from the first-stage design matrix by column name; no reliance on the call environment, so the extractor works after the model goes out of scope. The `ivreg` method uses `object$y`, `object$endogenous`, and `object$instruments` directly. Both error cleanly on non-IV models and on models with more than one endogenous variable.

3.4 The iv_check wrapper

`iv_check` inspects the model structure, detects which tests are applicable (binary versus discrete treatment, number of instrument levels, presence and dimensionality of exogenous controls), and runs only the applicable ones. The return is an `iv_check` object containing a tidy table with one row per test (name, statistic, p-value, verdict) plus an overall verdict string. When a user requests a specific test that is not applicable, `iv_check` warns and skips rather than silently dropping the request. In particular, `iv_kitagawa` is strictly the unconditional Kitagawa (2015) test and is dropped from the applicable list whenever the fitted model carries any exogenous control; `iv_mw` is the conditional test and supports up to one covariate in the current version (multivariate via tensor-product series basis is planned for v0.2.0). Automatic detection of the judge design is restricted to binary treatment in this version; multivalued judge designs should call `iv_testjfe` directly.

3.5 Bootstrap machinery

All three tests share a multiplier-bootstrap core. Rademacher weights are the default; a Mammen two-point option is available. The bootstrap process for each test is a linear transformation of the centred indicator or residual process, computed once per bootstrap replication via matrix-vector operations. Cell cap of 2 cores under `_R_CHECK_LIMIT_CORES_` keeps the package within CRAN example policy; interactive use defaults to `min(4, detectCores() - 1)`.

4 The Kitagawa (2015) test

4.1 Statistic and bootstrap

`iv_kitagawa` implements the variance-weighted Kolmogorov-Smirnov statistic of Kitagawa (2015) equation 2.1. For each pair `(z_L, z_H)` of instrument levels pre-ordered ascending by first-stage `E_hat[D | Z]`, and for each `d in {0, 1}`, the testable null implies that the interval-probability difference `P([y, y'], d | Z = z_L) - P([y, y'], d | Z = z_H)` is non-positive for every `y <= y'` (with the sign reversed for `d = 0`). Kitagawa's studentised statistic is

$$T_n = \sqrt{\tfrac{n_L n_H}{n_L+n_H}} \max_{(z_L,z_H,d)} \sup_{y\leq y'} \frac{\left[\Delta F_{\text{hat}}(y,y';d,z_L,z_H)\right]^+}{\hat\sigma(y,y';d,z_L,z_H)\vee\xi}, \tag{1}$$

where `n_L` and `n_H` are the sample sizes in the two instrument cells (so the pair-scale is the harmonic-mean form `sqrt(n_L n_H / (n_L + n_H))`), `Delta F_hat` is the signed interval-probability difference, `sigma_hat` is the plug-in binomial-mixture standard error with mixture weights matching Kitagawa's equation 2.1, and `xi` (the `se_floor` argument) is a small positive trimming constant that floors the denominator on intervals with vanishing estimated variance. Pairs are pre-ordered so the inequalities are one-sided and the statistic is strictly non-negative. Critical values come from a multiplier bootstrap: Rademacher weights `W_i in {-1, +1}` are applied to the centred indicator process `1{Y_i <= y, D_i = d} - \hat F(y, d | Z_i)` within each Z cell, the bootstrap analogue of `Delta F_hat` is recomputed, and it is studentised with the data-derived plug-in standard error frozen at the observed-sample value. Kitagawa's Algorithm 2.1 (Section 2.2) uses a pair-resampling bootstrap with replacement from the pooled empirical `H_N = \hat\lambda P_m + (1-\hat\lambda) Q_n`; the multiplier variant implemented here is asymptotically equivalent for this class of studentised functionals and is adopted for computational simplicity. The bootstrap p-value is the fraction of replications whose statistic exceeds `T_n`.

4.2 Implementation

The supremum over `(y, y')` is computed on a quantile grid of observed outcomes (default 50 points). The empirical-process sup is attained at sample `Y` values with probability one, so the quantile grid is a dense approximation to the exact sup up to Monte Carlo error. The `weighting` argument selects the variance-weighted form of Kitagawa's equation 2.1 (default) or the unweighted form of equation 2.2

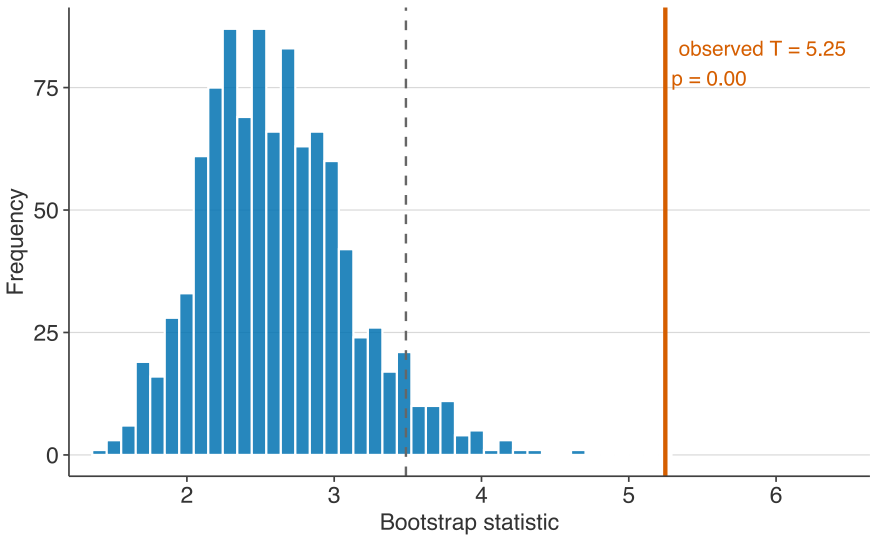

Figure 1: Kitagawa (2015) bootstrap distribution for the Card (1995) proximity-to-college IV, N = 3003. Observed test statistic (solid line) and the 95$^{\text{th}}$ percentile of the multiplier bootstrap (dashed). Treatment is a binary completed-college indicator (`educ >= 16`); the instrument is `near_college`. The observed statistic lies in the upper tail of the bootstrap distribution; the interval-sup test rejects the binary-discretised IV's testable implications at the 5% level. This should not be interpreted as a rejection of Card's original IV, which targets continuous years of schooling. The binary threshold at `educ >= 16` creates mixed complier subpopulations that make the testable implications bite.

(`weighting = "unweighted"`). The two are asymptotically equivalent under the null boundary; the weighted form has better finite-sample power when instrument cells have unequal sizes.

4.3 Card (1995) illustration

The bundled `card1995` dataset is a cleaned extract of 3,003 observations from the Card (1995) proximity-to-college IV for returns to schooling, with a binary `college` indicator added (`educ >= 16`). Figure 1 shows the bootstrap distribution of the Kitagawa statistic on this data using the `college` indicator as the binary treatment and `near_college` as the instrument.

5 The Mourifié-Wan (2017) test

5.1 Series regression with adaptive moment selection

For unconditional use without covariates, `iv_mw` delegates to the Kitagawa core with variance weighting; the two tests are asymptotically equivalent in that case and agree numerically under identical random seeds. The non-trivial implementation is the conditional path.

With a covariate vector `x`, `iv_mw` estimates the conditional CDF `F(y, d | X = x, Z = z)` by cubic-polynomial series regression of the indicator `1{Y <= y, D = d}` on a basis of `X`, restricted to observations at each level of `Z`. Standard errors are the heteroscedasticity-consistent sandwich `b(x)' (B'B)^{-1} (B' \text{diag}(r^2) B) (B'B)^{-1} b(x)` where `b(x)` is the basis evaluated at the target `x`. The studentised positive-part statistic is supped over a grid of `(y, x)` quantile points.

Critical values come from a multiplier bootstrap with Andrews-Soares (2010) adaptive moment selection. Let `xi_hat_k` denote the observed studentised statistic at index `k`. The bootstrap sup is restricted to the contact set `S = { k : xi_hat_k >= -kappa_n }` where `kappa_n = sqrt(log(log(n)))`. Moments with `xi_hat_k` well below the boundary are treated as strictly non-binding and dropped, giving tighter critical values than the plug-in least-favourable path without compromising size. The `sqrt(log(log(n)))` rate is deliberately small, on the slow end of the admissible range for this tuning sequence; the conditional-path size simulation below confirms that it controls size at the sample sizes considered.

5.2 Implementation surface

The user-facing arguments control the grid density (`x_grid_size`, `y_grid_size`), the basis (`basis_order`), and whether adaptive selection is applied (`adaptive = TRUE` by default). Bootstrap defaults (`n_boot = 1000`) deliver publication-grade p-values at `alpha = 0.05`.

6 The Frandsen-Lefgren-Leslie (2023) test

6.1 Statistic for binary treatment

`iv_testjfe` implements the asymptotic form of the Frandsen-Lefgren-Leslie (2023) test for designs with a binary treatment and an instrument consisting of `K` mutually exclusive dummy variables (the leniency-of-assigned-judge design). Under the joint null, the Wald estimator `(mu_j - mu_k) / (p_j - p_k)` identifies the same complier LATE for every pair of judges `(j, k)`, where `mu_j = E[Y | J = j]` and `p_j = E[D | J = j]`. Under binary treatment, the overidentification test that all pairwise LATEs agree is algebraically the weighted-least-squares test that `mu_j = alpha + beta * p_j` holds, divided by a pooled structural-residual variance estimator:

$$T_n = \frac{\sum_j n_j \left(\mu_j - \hat{\alpha} - \hat{\beta}\, p_j\right)^2}{\hat{\sigma}^2}, \qquad \hat{\sigma}^2 = \frac{1}{n-K} \sum_j \sum_{i:J_i=j} (y_i - \hat{\alpha} - \hat{\beta} d_i - \bar{u}_j)^2, \tag{2}$$

where `bar u_j` is the within-judge mean of the structural residuals and the reference distribution is `chi^2_{K - 2}`. Using structural residuals rather than within-judge variance of `y` removes the first-stage binomial contribution of `D`: the conditional variance decomposes as `Var(Y | J) = beta^2 p_j (1 - p_j) + sigma^2`, so the direct-y estimator inflates `sigma^2` by a factor of `1 + (beta / sigma)^2 p_j (1 - p_j)` relative to the correct denominator.

The `method = "bootstrap"` option replaces the chi-squared reference with a multiplier bootstrap of the restricted-residual process for small-`K` robustness. This is a simplification of the full Frandsen et al. (2023) test, which uses a flexible polynomial or spline specification for the outcome-propensity relationship and the Andrews-Soares (2010) moment-inequality procedure for the bounded-slope restriction. The chi-squared linearity test implemented here is the special case of their asymptotic form with `phi(p) = alpha + beta * p` (linear specification) and without the bounded-slope constraint.

6.2 Multivalued treatment

Frandsen, Lefgren, and Leslie (2023) state their test for binary `D`. We implement a direct generalisation to discrete `D` with `M + 1` distinct values: the scalar propensity `p_j` becomes the `M`-vector `(P(D = 1 | J), ..., P(D = M | J))`, the fit is the multiple WLS regression of `mu_j` on that vector, and the reference distribution is `chi^2_{K - M - 1}`. The same necessary-condition logic applies (each pair of judges identifies a common LATE vector under the joint null), and the degrees-of-freedom adjustment follows from the number of free parameters in the fit. `iv_testjfe` supports both binary and multivalued treatments; the extension requires no additional user input. We are not aware of a published proof covering the multivalued case for this particular statistic; users treating it as a published result should rely on the binary form.

The Kitagawa (2015) and Mourifié-Wan (2017) tests have an analogous extension to multivalued treatment via Sun (2023), who generalises the stochastic-dominance inequalities to cumulative-tail events `P(Y <= y, D <= d | Z)` and `P(Y <= y, D >= d | Z)`. `iv_kitagawa` implements this extension transparently: when the treatment vector has more than two distinct values, the test label switches from "Kitagawa (2015)" to "Sun (2023)" and the statistic aggregates violations across `2M` inequality families instead of `2`. Sun's asymptotic validity and bootstrap procedure carry through from the binary case.

6.3 Diagnostic output

The returned object includes the `K x K` matrix `pairwise_late` of pairwise Wald estimates (binary case only), and `worst_pair`, the judge pair whose Wald LATE deviates most from the fitted slope. These are diagnostic in the sense of the paper's figures: a pair whose Wald LATE is far from the common slope is the first place to look when investigating a rejection.

7 Case study: Card (1995) proximity to a four-year college

The Card (1995) IV estimate of the return to schooling is the canonical applied IV illustration. The unconditional Kitagawa / Mourifié-Wan tests reject the binary discretisation of Card's continuous schooling variable; once a single demographic control is included, the conditional Mourifié-Wan test does not reject. The contrast is the right reading of Card's design: proximity-to-college is a plausible instrument only conditional on demographic and regional controls.

```
library(fixest)
library(ivcheck)

data(card1995)

# Unconditional: both tests reject.
m_uncond <- feols(lwage ~ 1 | college ~ near_college, data = card1995)
iv_check(m_uncond, n_boot = 500)
#> Kitagawa (2015):     stat = 5.25, p = 0.00, reject
#> Mourifie-Wan (2017): stat = 5.25, p = 0.00, reject

# Conditional on age: conditional Mourifie-Wan does not reject.
m_cond <- feols(lwage ~ age | college ~ near_college, data = card1995)
iv_check(m_cond, n_boot = 200)
#> i Kitagawa skipped (model has controls; iv_kitagawa is unconditional).
#> Mourifie-Wan (2017): stat = 79.5, p = 0.71, pass
```

The wrapper detects that `college` is binary and `near_college` is a binary instrument. In the unconditional specification it runs both Kitagawa and Mourifié-Wan (which coincide without covariates) and reports rejection. In the conditional specification it drops Kitagawa from the applicable list (with an informational message) and runs the conditional Mourifié-Wan test alone; the conditional CLR series-regression form does not reject. The two readings are not in conflict: the unconditional test catches a violation that is removed once age is conditioned on. Card's canonical specification uses additional controls (race, region); multivariate conditioning via tensor-product series basis is on the v0.2.0 roadmap, and the v0.1.2 workaround is dimension reduction to a single propensity index.

Users running the test on their own binary-IV designs should inspect the `binding` element of the result to see which outcome interval carries the violation, and consider whether the binary discretisation itself is driving the finding. This is a test of the binary `educ >= 16` threshold, not Card's original continuous-D IV.

8 Monte Carlo validation

8.1 Null distribution of `iv_testjfe`

Figure 2 shows the empirical distribution of `T_n` from equation (2) over 1000 Monte Carlo replications under a valid-IV null, overlaid with the asymptotic chi-squared density with `K - 2 = 18` degrees of freedom. Empirical moments are 17.67 (target 18.0) and 33.44 (target 36.0); the 95$^{\text{th}}$ percentile is 28.28 against the chi-squared critical value of 28.869. Empirical size at nominal 5% is 3.9% with the asymptotic method and 4.3% with `method = "bootstrap"`, against a Monte Carlo standard error of 0.69 percentage points; both methods sit within two MC standard errors of nominal and are mildly conservative at this sample size. Two mechanisms contribute to the conservatism. First, the within-judge residual-variance estimator in equation (2) absorbs `K` degrees of freedom, which at finite `K` over-estimates the structural variance assumed by the asymptotic theory. Second, the per-judge propensities `\hat p_j` enter the fit as estimated regressors, and their sampling covariance with `\hat \mu_j` is not accounted for in the chi-squared reference, which treats `\hat p_j` as fixed. Both effects shrink as `n_j` grows; the bootstrap approximates the true sampling distribution directly and is recommended for publication-grade p-values at modest `n_j`. The asymptotic chi-squared remains accurate in the upper tail for diagnostic purposes.

8.2 Null size under finite samples and skewed Z

Monte Carlo size simulations at nominal 5% across 24 configurations (sample sizes 300, 800, 2000; first-stage propensities `P(D=1|Z_low), P(D=1|Z_high) \in \{(0.5, 0.5), (0.3, 0.7), (0.3, 0.4), (0.1, 0.9)\}`; Z-cell balance 50/50 or 35/65; 500 Monte Carlo replications per cell) drove the choice of the default `se_floor` trimming constant (Kitagawa's `\xi`). Kitagawa (2015, p.~2051) informally

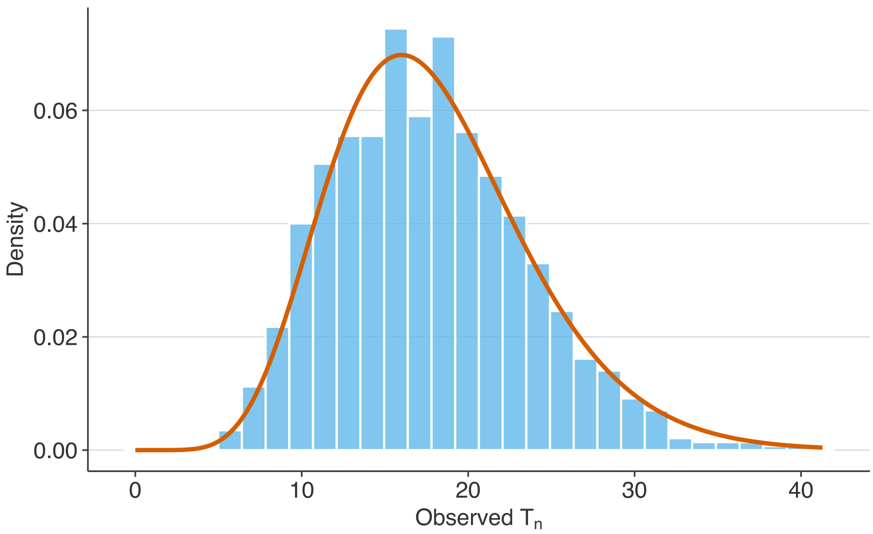

Figure 2: Monte Carlo null distribution of the Frandsen-Lefgren-Leslie test statistic, K = 20 judges and N = 3000 observations per replication. Histogram shows the empirical density from 1000 replications under the null (Y independent of J given D). Red curve is the asymptotic chi-squared distribution with K - 2 = 18 degrees of freedom implied by equation (2). Agreement in mean and upper-tail quantile is close; the structural-residual variance estimator and the estimated-regressor adjustment for $\hat{p}_j$ (not reflected in the chi-squared reference) produce a mildly conservative test.

recommends `\xi \in [0.05, 0.1]`. At the lower end of that range the test was anti-conservative (5-11% empirical rejection at nominal 5%) under skewed Z-cell distributions with weak first stages at sample sizes below 1500. Raising the floor to `\xi = 0.15` brings empirical size at or below nominal 5% in 22 of the 24 configurations; the two cells that exceed nominal are both at `n = 300` with skewed Z-cell balance (35/65), landing at 7.2% (balanced first stage `P(D=1|Z) = 0.5`) and 6.0% (card-like `P(D=1|Z) \in \{0.3, 0.4\}`) respectively. These lie approximately 2.3 and 1.0 Monte Carlo standard errors above nominal (`\sqrt{0.05 \times 0.95 / 500} \approx 0.97` percentage points) and decay monotonically with sample size: at `n = 800` the same cells land at 4.0% and 1.8%, at `n = 2000` at 1.6% and 0.6%. At the practical sample sizes for Card (1995) and similar designs (`n \geq 800`) empirical size is at or below nominal throughout. The `se_floor` argument is user-tunable; setting `se_floor = 0.1` matches the upper end of Kitagawa's recommended range for users who want to stay inside it.

8.3 Size of the conditional Mourifié-Wan test

The conditional series-regression path of `iv_mw` is validated separately, since it is the only component not shared with the Kitagawa core. Under a valid-IV null with a single continuous covariate (the treatment propensity monotone in `Z`, the outcome depending on the treatment and the covariate but not directly on `Z`), the empirical size at nominal 5% is 4.0% at `n = 800` and 4.8% at `n = 2000`, over 500 Monte Carlo replications each (Monte Carlo standard error approximately 0.97 percentage points). The Chernozhukov-Lee-Rosen intersection-bounds inference with Andrews-Soares moment selection at `kappa_n = sqrt(log(log(n)))` therefore controls size on the conditional path at the sample sizes typical of applied work.

8.4 Power under exclusion violation

Figure 3 shows the empirical rejection rate of `iv_kitagawa` under a D-specific direct effect of `Z` on `Y`, the class of exclusion violation the Kolmogorov-Smirnov test is designed to detect. At `delta = 0` (null) the empirical size is roughly nominal; by `delta = 1.5` in units of `sigma`, power approaches one. Figure 4 shows the analogous curve for `iv_testjfe` under a sinusoidal direct judge effect, the leading violation mode for the judge-IV design; the judge-design test is highly powered at these sample sizes, with rejection rising from nominal at `eta = 0` to essentially one by `eta = 0.2` (in units of `sigma`).

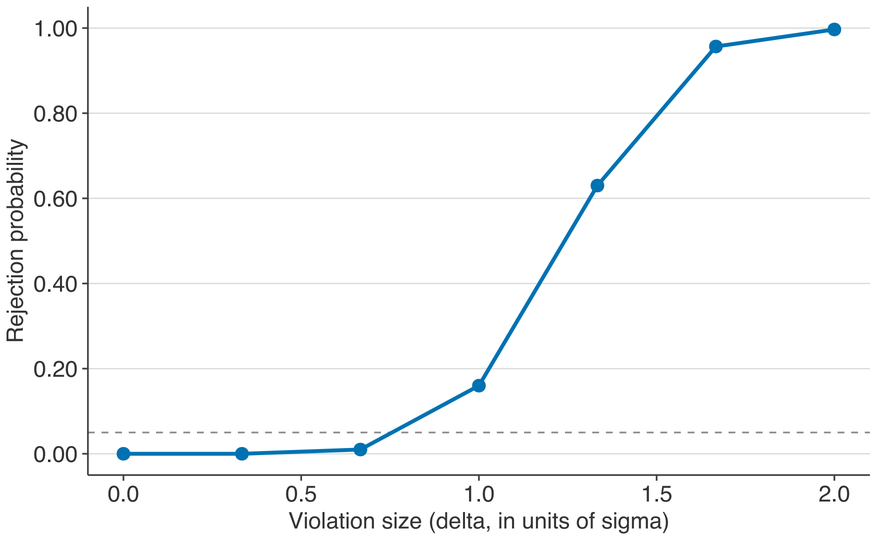

Figure 3: Empirical power of `iv_kitagawa` under a D-specific direct-effect exclusion violation, N = 500 per replication. Horizontal dashed line shows the nominal 5% level. The simulator generates data with $Y = D + \delta\sigma \cdot D \cdot (Z - z_L) + \varepsilon$, directly violating the testable inequality for the $d = 1$ cells. Power approaches one by $\delta = 1.5\sigma$ with 300 Monte Carlo replications per grid point.

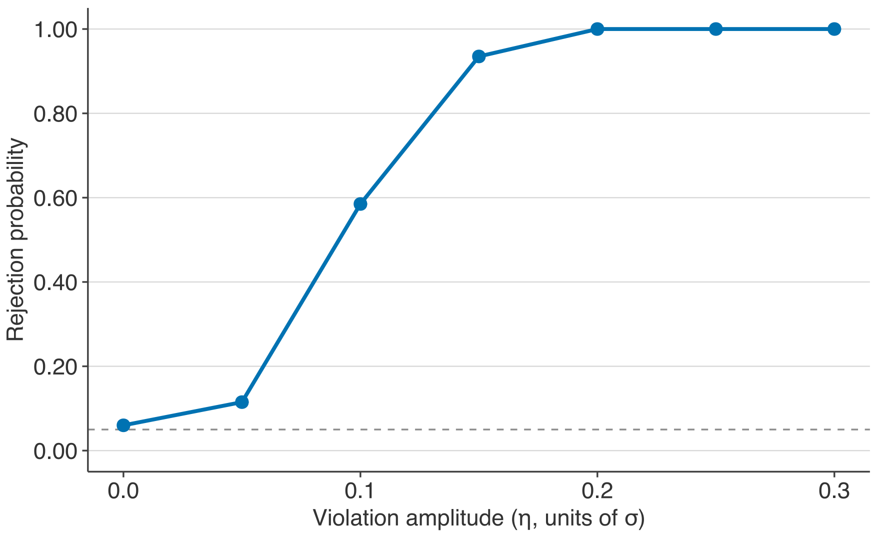

Figure 4: Empirical power of `iv_testjfe` under a sinusoidal judge-effect exclusion violation, K = 20 judges and N = 3000. Horizontal dashed line shows the nominal 5% level. The simulator generates data with $Y = D + \eta\sigma \sin(0.5J) + \varepsilon$, a direct effect of judge identity on Y holding D fixed. Power rises steeply in the violation amplitude η and reaches essentially one by $\eta = 0.2\sigma$, with 200 Monte Carlo replications per grid point; the asymptotic chi-squared p-value is used throughout. The test is highly powered against this violation mode at $K = 20$, $N = 3000$.

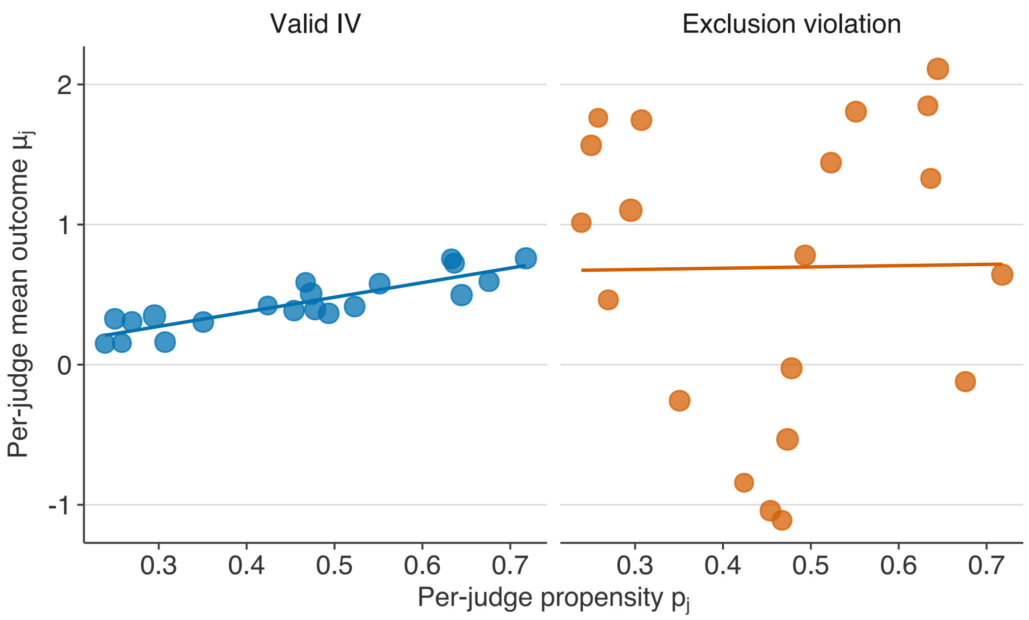

Figure 5: Per-judge outcome means versus per-judge treatment propensities, K = 20 judges and N = 3000. Left: valid-IV null ($Y = D + \varepsilon$). Right: exclusion violation with direct judge effect $1.5 \sin(0.5J)$. Point size is proportional to judge-cell sample size; lines are weighted-LS fits. The violation design produces a systematic departure from linearity that `iv_testjfe` detects at $p < 10^{-4}$.

8.5 Judge-IV diagnostic figures

Figure 5 plots per-judge outcome means against per-judge propensities for a valid-IV simulation and an exclusion violation. Under validity the points fit the regression line; under a violation they deviate in a pattern related to the direct judge effect.

Figure 6 shows the pairwise Wald LATE matrix for the violation design. Under validity, every off-diagonal entry estimates the common complier LATE; under the violation, pairs involving specific judges diverge markedly. `iv_testjfe()$worst_pair` surfaces the pair with the largest deviation from the fitted slope.

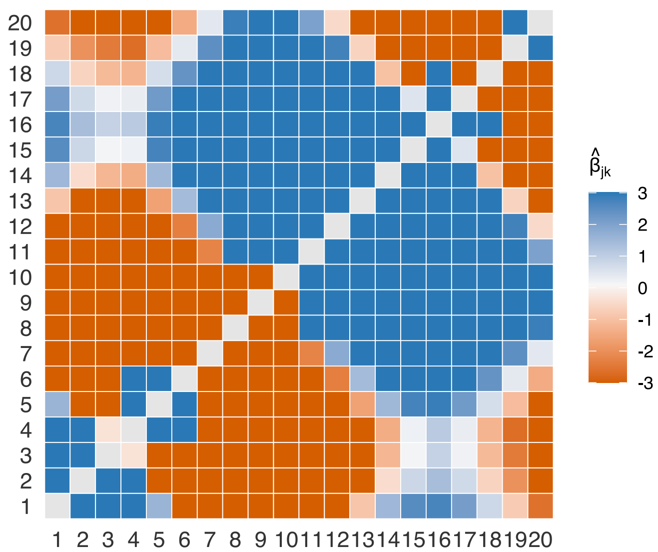

Figure 6: Pairwise Wald LATE matrix $(\mu_j - \mu_k)/(p_j - p_k)$ under the violation design from Figure 5. Entries are colour-shaded around the fitted slope; deviations in either direction signal pairs whose Wald estimates are inconsistent with a common complier LATE. The matrix is returned as `pairwise_late` on the `iv_test` object; the worst-offender pair is also returned as `worst_pair` for direct inspection.

9 Limitations

The following limitations apply.

1. **ivcheck** is a testing package, not an estimator. Use **fixest** or **ivreg** for the IV estimation itself; **ivcheck** operates on the fitted model or on raw vectors.
2. The current version supports discrete instruments only. Continuous `Z` must be discretised into bins (quartiles or quintiles) before passing to `iv_kitagawa` or `iv_mw`, at modest cost in power. A nonparametric continuous-`Z` extension is on the v0.2.0 roadmap.
3. Fuzzy regression discontinuity has a dedicated test (Arai et al., 2022) that is not yet in **ivcheck**. Its different infrastructure (running variable, cutoff, local-polynomial bandwidth, bias correction) does not fit the current spine of S3 dispatch on fitted IV models, so a dedicated `iv_frd()` function is deferred to v0.2.0.
4. `iv_testjfe` implements the asymptotic chi-squared linearity test that corresponds to the Frandsen et al. (2023) statistic under a linear outcome-propensity specification. The full flexible-series test with the bounded-slope moment-inequality restriction is planned for v0.2.0. Users needing the published test with the flexible basis should run Frandsen's Stata `testjfe` module in the interim.
5. Fixed-effects IV models are not yet supported. Calling `iv_kitagawa`, `iv_mw`, or `iv_testjfe` on a `fixest` model with the `| FE |` component aborts with a clear error. The discrete-Z tests operate on the raw `(Y, D, Z)` joint distribution, and residualising on fixed effects destroys the discrete structure of `Z`. For judge-style FE designs the right tool is `iv_testjfe` directly (the FE *are* the instrument variation); for non-judge FE the documented workaround is to pre-demean `Y` and `D` within each FE cell, keep `Z` discrete, and call the default method on the resulting vectors.
6. `iv_kitagawa` is strictly the unconditional Kitagawa (2015) test. Dispatched on a fitted `fixest` or `ivreg` model that carries exogenous controls, it errors with a pointer to `iv_mw` for the conditional Mourifié-Wan (2017) test, rather than silently dropping the controls. (Through the `iv_check` wrapper the same condition instead causes Kitagawa to be dropped from the applicable list with an informational message, rather than an error.) Users wanting the unconditional test on data that happen to come from a controlled regression should call `iv_kitagawa` on raw `(y, d, z)` vectors.
7. The conditional path of `iv_mw` supports a single covariate. Users passing a matrix of multiple covariates receive a clear error pointing to the tensor-product-basis extension planned for v0.2.0. The current recommendation is to condition on the covariate most plausibly driving heterogeneity in compliance, or to reduce a multivariate `X` to a single propensity index.
8. Rejection of any of the three tests is evidence of violation but does not localise the failed assumption. Non-rejection is evidence of no detectable violation at level `alpha`, not proof of

validity. Report non-rejection as such.

9.1 Notes on fidelity

For transparency, the ordered multivalued `D` path in `iv_kitagawa` tests a richer family of inequalities than Sun (2023) equation 10: we sweep over cumulative-tail events `P(Y <= y, D <= l | Z)` and `P(Y <= y, D >= l | Z)` for every intermediate `l` and every `Z` pair, while Sun's derivation uses only `d_min` and `d_max` across adjacent pairs. All inequalities tested hold under Sun's Assumption 2.2, so the test controls size; the expanded family gives a more exhaustive test rather than the minimal-implication version of the paper. The unordered case (Sun section 3.3) is available via `treatment_order = "unordered"` with a user-specified `monotonicity_set` encoding the direction of the monotonicity restriction.

`iv_testjfe` implements the linearity-test form of the Frandsen et al. (2023) overidentification test: we fit `mu_j` against `p_j` by WLS and compare the weighted residual SS to a `chi^2_{K-2}` reference. FLL's published test uses a flexible basis plus bounded-slope moment-inequality constraints on the LATE; our v0.1.0 form is a simpler necessary-condition check. Users wanting the exact published test should run Frandsen's Stata `testjfe` module; a flexible-basis extension and CLR-style bounded-slope inference are planned for v0.2.0.

The `iv_mw` CLR path is an independent implementation of the Chernozhukov et al. (2013) intersection-bounds framework (series-regression conditional CDF, robust plug-in SE, Andrews and Soares (2010) adaptive moment selection) rather than a port of Mourifié-Wan's Matlab code. Cross-validation against the authors' code is on the v0.2.0 wishlist. The bootstrap multiplier defaults to Rademacher but `multiplier = "gaussian"` and `multiplier = "mammen"` are available for robustness checks.

10 Conclusion

ivcheck closes the tooling gap between the econometric literature on LATE falsification and applied IV practice: three published falsification tests, each a single function call on a fitted **fixest** or **ivreg** model, with a wrapper that selects the applicable tests automatically. Monte Carlo validation shows empirical size at or below nominal across the tested designs, and power approaching one by a direct-effect violation of roughly 1.5σ for the Kitagawa test. Planned extensions for v0.2.0 include non-parametric continuous-instrument inference, multivariate conditioning via a tensor-product series basis, the full flexible-basis Frandsen-Lefgren-Leslie test with bounded-slope inference, and a fuzzy regression-discontinuity test (`iv_frd()`) following Arai et al. (2022). The package is available on CRAN; source and issue tracker at `https://github.com/charlescoverdale/ivcheck`.

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